1. Introduction

In air layering (marcotting), success is not only about technique—it is also about **precision, cleanliness, and the quality of tools used**. Poor tools or contamination can lead to infection, rot, or total failure of the propagation process.

This module focuses on:

- Professional-grade tools and their proper use
- Selecting high-quality materials
- Sterilization techniques to prevent disease

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2. Essential Tools: Beyond the Basics

While Module 01 introduced basic tools, here we refine your toolkit for **maximum efficiency and success**.

Primary Cutting Tools

Pruning Knife

- Sharp, thin blade for precise girdling
- Ideal for controlled bark removal

Pruning Shears (Secateurs)

- Used for cutting the branch after rooting
- Should be sharp to avoid crushing plant tissue

Advanced Tip

A dull blade can:

- Tear the bark unevenly
- Damage the cambium improperly
- Increase infection risk

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3. Wrapping and Sealing Materials

Plastic Wrap (Polyethylene Film)

- Retains moisture
- Transparent—allows root monitoring

Aluminum Foil

- Reflects sunlight

- Prevents overheating
- Creates a dark environment for roots

Twine, ties, or Electrical Tape

- Secures both ends tightly
- Prevents moisture loss

Professional Insight

Double-wrapping (plastic + foil) significantly improves success rates in hot climates.

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4. Rooting Media: Advanced Selection Criteria

Not all media are equal. The ideal rooting medium must balance:

- **Moisture retention**
- **Aeration (oxygen availability)**
- **Sterility**

Top Choices

Sphagnum Moss

- High water retention
- Naturally anti-fungal
- Preferred by professionals

Coconut Coir

- Sustainable alternative
- Excellent aeration
- Locally accessible

Optional Additions

- Perlite (for improved aeration)
- Vermiculite (for moisture balance)

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5. Sterilization: Why It Is Critical

When you girdle a branch, you create an **open wound**. Without sterilization, this becomes an entry point for:

- Bacteria

- Fungi
- Viruses

Result of Poor Sterilization

- Rotting of the branch
- Failed root formation
- Spread of disease to the mother plant

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6. Sterilizing Your Tools

Method 1: Alcohol (Recommended)

- Use 70% isopropyl alcohol
- Wipe or dip blade before and after each cut

Method 2: Flame Sterilization

- Pass blade through flame briefly
- Allow it to cool before use

Method 3: Bleach Solution

- Mix 1 part bleach with 9 parts water
- Soak tools for a few minutes

Best Practice

Always sterilize:

- Before starting
- Between different plants
- After completing work

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7. Preparing the Rooting Medium

Even if your medium looks clean, it may contain microorganisms.

Preparation Steps

1. Soak medium in clean water
2. Squeeze out excess water (should be moist, not dripping)
3. Optionally rinse with mild fungicide solution

Consistency Check

- When squeezed: no dripping water
- When released: retains shape

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8. Environmental Hygiene

Clean tools alone are not enough—your working environment matters.

Best Practices

- Work in a shaded, clean area
- Avoid working during rain (high contamination risk)
- Keep materials off the ground
- Use clean containers for soaking media

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9. Common Mistakes with Tools and Materials

Mistake 1: Using Rusty Tools

- Introduces pathogens
- Causes uneven cuts

Mistake 2: Overly Wet Medium

- Leads to root rot

Mistake 3: Loose Wrapping

- Causes moisture loss
- Delays root formation

Mistake 4: Skipping Sterilization

- One of the top causes of failure

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10. Demonstration Scenario

Example Workflow (Professional Setup):

1. Sterilize pruning knife with alcohol
2. Prepare moist sphagnum moss
3. Perform girdling with a clean cut
4. Apply rooting hormone
5. Wrap with plastic

6. Cover with aluminum foil
7. Seal tightly with tape

This systematic approach ensures:

- Minimal contamination
- Optimal moisture retention
- Faster root development

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11. Module Summary

By the end of this module, you should be able to:

- Identify and use professional-grade tools
- Select the best rooting materials
- Apply proper sterilization techniques
- Prepare a clean and controlled propagation setup